Curriculum Vitae

# YUHAO JIANG

Postdoctoral Researcher, EPFL

## CONTACT

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## RESEARCH PROFILE

My research advances reconfigurable intelligence by treating robot morphology as a decision variable in planning and control. I develop mechanically intelligent mechanisms that generate selectable motions with few actuators and laminate-based foldable robotic systems designed for scalable fabrication. Complementing this hardware research, I develop computational methods for multi-module coordination and physically grounded embodiment planning, enabling robots to assess and exploit the capabilities of different configurations. My long-term goal is to establish an integrated framework for reconfigurable physical AI, enabling robots to predict, select, and adapt their embodiments and behaviors to meet changing task and environmental demands. Beyond individual robots, these systems could generate diverse physical interaction data across robot embodiments and challenging environments, supporting broader advances in physical AI.

## EDUCATION

**Arizona State University, Tempe, USA** *Jan. 2019 – Aug. 2023*
Ph.D. in Mechanical Engineering
**Advisor:** Prof. Daniel Aukes
**Dissertation:** Design and Modeling of Soft Curved Reconfigurable Anisotropic Mechanisms

**University of Florida, Gainesville, USA** *Sept. 2015 – May 2017*
Master of Science in Mechanical Engineering

**Donghua University, Shanghai, China** *Sept. 2011 – Jun. 2015*
Bachelor of Engineering in Mechanical Engineering

## PROFESSIONAL EXPERIENCE

**Swiss Federal Institute of Technology Lausanne (EPFL), Switzerland** *Sept. 2023 – Present*
Postdoctoral Researcher, Reconfigurable Robotics Lab
**Supervisor:** Prof. Jamie Paik

## PUBLICATIONS

### Journal Publications

[1] **Y. Jiang**, F. Chen, J. Paik, and D. M. Aukes, "Vibration of Soft, Twisted Beams for Under-Actuated Quadrupedal Locomotion," IEEE/ASME Transactions on Mechatronics, April 2026, 10.1109/TMECH.2026.3686693.

[2] X. Zheng, **Y. Jiang**, M. Mete, J. Li, I. Watanabe, T. Yamada, and J. Paik, "Metamaterial Robotics," Science Robotics, 10, eadx1519 (2025). 10.1126/scirobotics.adx1519.

[3] **Y. Jiang**, S. Asmar, Z. Wang, S. Demirtas, and J. Paik, "CPG-based Manipulation with Multi-Module Origami Robot Surface," IEEE Robotics and Automation Letters, March 2025, 10.1109/LRA.2025.3555381.

[4] **Y. Jiang**, F. Chen and D. M. Aukes, "Tunable Dynamic Walking via Soft Twisted Beam Vibration," IEEE Robotics and Automation Letters, vol. 8, no. 4, pp. 1967–1974, April 2023, 10.1109/LRA.2023.3244716.

[5] M. Sharifzadeh, **Y. Jiang**, A. Lafmejani, K. Nichols, and D. M. Aukes, "Maneuverable gait selection for a novel fish-inspired robot using a CMA-ES-assisted workflow," Bioinspiration & Biomimetics, vol. 16, no. 5, pp. 056017, August 2021, 10.1088/1748-3190/ac165d.

[6] M. Sharifzadeh, **Y. Jiang**, and D. M. Aukes, "Reconfigurable Curved Beams for Selectable Swimming Gaits in an Underwater Robot," IEEE Robotics and Automation Letters, vol. 6, no. 2, pp. 3437–3444, April 2021, 10.1109/LRA.2021.3063961.

**Conference Publications**

[1] **Y. Jiang**, M. Sharifzadeh, and D. M. Aukes, "Shape Change Propagation Through Soft Curved Materials for Dynamically-Tuned Paddling Robots," 2021 IEEE 4th International Conference on Soft Robotics (RoboSoft), 2021, pp. 230–237, 10.1109/RoboSoft51838.2021.9479208.

[2] **Y. Jiang**, M. Sharifzadeh, and D. M. Aukes, "Reconfigurable Soft Flexure Hinges via Pinched Tubes," 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020, pp. 8843–8850, 10.1109/IROS45743.2020.9341109.

[3] D. Bailey, R. Moreno, S. Demirtas, Z. Wang, **Y. Jiang**, J. Paik, K. Stoy, and A. Faíña. "Flexible and Foldable: Workspace Analysis and Object Manipulation Using a Soft, Interconnected, Origami-Inspired Actuator Array." 2026 IEEE International Conference on Robotics and Automation (ICRA), 2026, Accepted, arxiv.org/abs/2509.13998.

[4] P. Bupe, **Y. Jiang**, J. Lin, T. Nguyen, M. Han, D. Aukes, C. Harnett, "Embedded Optical Waveguide Sensors for Dynamic Behavior Monitoring in Twisted-Beam Structures," 2024 IEEE 7th International Conference on Soft Robotics (RoboSoft), San Diego, CA, USA, 2024, pp. 139–144, 10.1109/RoboSoft60065.2024.10521938.

[5] M. Sharifzadeh[†], **Y. Jiang**[†], A. Lafmejani, D. M. Aukes, "Compensating for Material Deformation in Foldable Robots via Deep Learning – A Case Study," 2022 IEEE International Conference on Robotics and Automation (ICRA), 2022, 10.1109/ICRA46639.2022.9811752.

[6] M. Sharifzadeh, **Y. Jiang**, R. Khodambashi, D. M. Aukes, "Increasing the Life Span of Foldable Manipulators With Fabric." Proceedings of the ASME 2020 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. Volume 10: 44th Mechanisms and Robotics Conference (MR). Virtual, Online. August 17–19, 2020. V010T10A087. ASME, 10.1115/DETC2020-22757.

[†] Equal contribution.

## PATENTS

[1] "Pinched tubes for reconfigurable robots", D. Aukes, M. Sharifzadeh, **Y. Jiang**, N. Gravish, M. Jiang - US Patent US12017347B2

[2] "Buckling beams for underwater and terrestrial autonomous vehicles", D. Aukes, M. Sharifzadeh, **Y. Jiang** - US Patent US12240283B2

[3] "Mechanisms for steering robotic fish", D. Aukes, M. Sharifzadeh, K. Nichols, **Y. Jiang** - US Patent US11124281B2

[4] "Locomotion via vibration of soft, twisted beams with an under-actuated quadruped", D. Aukes, **Y. Jiang**, F Chen - US Patent Application US20260115895A1

[5] "Tunable Motion Using Flexible Twisted Beams", D. Aukes, **Y. Jiang**, F. Chen - US Patent Application 20240391542

## GRANT ACTIVITIES

**MusicTouch: Haptic Robotic Platform for Musical Experience**
*PI* | EPFL–UNIL CROSS 2027 | CHF 100,000 | Awarded *Sept. 2026 – Sept. 2027*

**COBRAS: Circular On-site Building with Robotic Autonomous Swarms**
*Co-applicant* | EU EIC Pathfinder | EUR 4 million | Awarded *Nov. 2026 – Nov. 2029*

## SELECTED PROJECTS

**MOZART: Morphing Computerized Mats with Embodied Sensing and Artificial Intelligence** *Sept. 2023 – Present*
EPFL, Reconfigurable Robotics Lab
**Funded by:** EU Horizon Europe Research and Innovation Programme. Grant number: 101069536.

**SCRAM: Soft Curved Reconfigurable Anisotropic Mechanisms** *Jan. 2020 – May 2023*
Arizona State University, IdeaLab
**Funded by:** NSF EFRI C3 SoRo. Grant number: 1935324.

**FIRE: Fish-Inspired Robots for Extreme Environments** *Jan. 2019 – Jan. 2020*
Arizona State University, IdeaLab
**Funded by:** Salt River Project

## TEACHING AND STUDENT MENTORING

### Course Instruction

**ME410: Mechanical Engineering Product Design and Development** *Fall 2023–2026*
*Mechanical Engineering, EPFL*

**ME420: Advanced Design for a Sustainable Future** *Fall 2024–2026*
*Mechanical Engineering, EPFL*

### Graduate Student Mentoring

**Gonçalo Pais** *M.Sc. Mechanical Engineering | Master's semester project* *Spring 2025*
Development of Energy Storage and Release Mechanisms for Rapid Dynamic Motions in Canfield Origami Robots

**Aurora Ruggeri** *M.Sc. Mechanical Engineering | Master's semester project* *Spring 2024*
Study of soft metamaterials for object sensing and geometry generation

**Louis Flahault** *M.Sc. Robotics | Master's semester project* *Spring 2024*
Kinematic study and design of a spatially reconfigurable modular robotic platform

**Serge Asmar** *M.Sc. Robotics | Master's semester project* *Spring 2024*
Locomotion design and control using surface-wave changes generated by the Ori-Pixel platform

**Nicolas Nouel** *M.Sc. Robotics | Master's thesis | Co-advisor* *Spring 2024*
Programmable surface using bistable structures

## TALKS

### Conference Proceedings Talks

[1] **IEEE-IROS 2025:** "CPG-based Manipulation with Multi-Module Origami Robot Surface"

[2] **IEEE-RoboSoft 2023:** "Tunable Dynamic Walking via Soft Twisted Beam Vibration"

[3] **IEEE-ICRA 2022:** "Compensating for Material Deformation in Foldable Robots Via Deep Learning – a Case Study", https://youtu.be/AwS4vabv-JQ

[4] **IEEE-ICRA 2021:** "Reconfigurable Curved Beams for Selectable Swimming Gaits in an Underwater Robot", https://youtu.be/EszTDc9slyw

[5] **IEEE-RoboSoft 2021:** "Shape Change Propagation Through Soft Curved Materials for Dynamically-Tuned Paddling Robots"

[6] **IEEE-IROS 2020:** "Reconfigurable Soft Flexure Hinges via Pinched Tubes", https://youtu.be/J5heXXD6mVo

### Workshop Presentations

[1] **IEEE-IROS 2026:** "Reconfigurable Robots for Programmable Physical Intelligence" (Invited talk)

[2] **IEEE-RoboSoft 2025:** "CPG-Based Motion Generator for Multi-Module Origami Robot"

[3] **IEEE-RoboSoft 2023:** "Model Order Reduction for Vibrational Soft Twisted Beams Using Pseudo-rigid-body Modeling – A Case Study" (**awarded 2nd place for best talk**)

[4] **IEEE-ICRA 2022:** "Modular Robots Using Soft Curved Reconfigurable Anisotropic Mechanisms"

### Seminar Talks

[1] "Empowering Actuation of Soft Robotic Systems via Soft Curved Reconfigurable Anisotropic Mechanism", hosted by Prof. Nick Gravish and Prof. Michael Tolley, UCSD, Feb. 2023.

## ACADEMIC SERVICE

### Event Organization

[1] **IEEE-IROS 2025 Workshop :** "S'MORE: Shape-Morphing Robotics via Embodied Sensing and Mechanisms", https://www.shapemorphingrobot.com/

[2] **2024 RRL Demo Day:** Full-day public event for research projects from RRL and ME-410 and ME-420 class,
https://www.paikslab.com/courses

[3] **2023 RRL Demo Day:** Full-day public event for research projects from RRL and ME-410 class, https://www.paikslab.com/courses

[4] **Robosoft 2021 Workshop :** "Breaking the Mold: Challenging Current Paradigms in Soft Robotics", https://www.scrambots.com/robosoft-2021-workshop

### Media Interview

[1] **RTS Education and Scientific Program:** feature in "A guide to the future: Swiss Federal Institute of Technology 02", https://www.youtube.com/watch?v=9yoNLg5Qho0&t=560s

**Editor**

IEEE-IROS 2026: Associate Editor

**Journal Reviewer**

The International Journal of Robotics Research (IJRR)
IEEE Transactions on Robotics (T-RO)
IEEE/ASME Transactions on Mechatronics (T-Mech)
IEEE Robotics and Automation Letters (RA-L)
Soft Robotics (SoRo)
Journal of Field Robotics (JFR)
ASME Journal of Mechanisms and Robotics (JMR)

**Conference Reviewer**

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
International Conference on Robotics and Automation (ICRA)
International Conference on Soft Robotics (Robosoft)
ACM Symposium on Computational Fabrication (SCF)

**Grant Reviewer**

Swiss National AI Institute (Seed Grant Program)

**Demos and Expositions**

[1] RRL lab tours ($\sim$ 6 times per year)

[2] Swiss Robotics Day (2023–2025)